

Suggested FYP topics for 2026-27

by

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Project code :	YZ1
Project title :	Empirical Study on USDT On-Chain Freezing Behavior and Intelligence Analysis
No. of students :	2
Description of project :	Tether, the issuer of USDT, holds the centralized authority to freeze on-chain addresses via smart contract functions. Hundreds of addresses holding significant assets have been blacklisted to date, yet there is limited systematic research on the patterns, rationale, and effectiveness of these freezing actions. This project conducts an empirical study by extracting all USDT blacklist events from Ethereum and Tron, analyzing the transaction behavior and inter-relationships of frozen addresses, and cross-referencing with open-source intelligence such as OFAC sanctions lists, known hack timelines, and law enforcement announcements. The project will deliver a research paper with findings on freezing patterns and trends, a data analysis toolkit with visualization dashboard, and a curated dataset of frozen addresses with associated on-chain intelligence.
Industrial collaboration :	No.

Project code :	YZ2
Project title :	AI-Augmented Regulatory Compliance Assistant for Virtual Asset Service Providers
No. of students :	1-2
Description of project :	Virtual asset service providers in Hong Kong operate under an increasingly complex regulatory landscape, including the Anti-Money Laundering and Counter-Terrorist Financing Ordinance (AMLO), Securities and Futures Commission (SFC) guidelines, Hong Kong Monetary Authority (HKMA) circulars, and Financial Action Task Force (FATF) recommendations. Compliance teams often struggle to efficiently retrieve and interpret relevant provisions across these overlapping frameworks. This project builds a Retrieval-Augmented Generation (RAG) based question-answering system that allows compliance professionals to query regulatory requirements in natural language, with responses grounded in and citing the original regulatory text. The system will support document upload, cross-jurisdictional rule comparison, and a web-based interface for practical daily use.
Industrial collaboration :	No.

Project code :	YZ3
Project title :	Empirical Study on the OTC Crypto Trading Landscape in Hong Kong: Compliance, Risks, and Regulatory Gaps
No. of students :	2

Description of project :	Over-the-counter (OTC) crypto trading is a significant yet under-studied segment of Hong Kong's virtual asset ecosystem. Physical OTC shops are concentrated in areas such as Mong Kok and Wan Chai, while numerous online OTC desks and peer-to-peer platforms operate with varying degrees of regulatory oversight. Following the implementation of the VASP licensing regime under AMLO, the compliance status of these OTC operators remains unclear, and their role in facilitating potential money laundering, fraud proceeds, and sanctions evasion has drawn increasing attention from law enforcement. This project conducts a systematic empirical investigation of Hong Kong's OTC crypto trading landscape. The study will map the current ecosystem through public data collection including business registry records, online platform surveys, and open-source intelligence, identify the typologies of OTC-related illicit finance risks such as cash-to-crypto conversion for fraud proceeds and trade-based money laundering, analyze on-chain transaction patterns associated with known OTC addresses to assess fund flow characteristics, and examine the regulatory gaps between the current VASP licensing framework and the operational reality of OTC services. The project will deliver a research report with an OTC ecosystem mapping, a risk typology framework, on-chain analysis findings, and policy recommendations for Hong Kong regulators on closing the OTC compliance gap.
Industrial collaboration :	Yes, With BlockSec.

Project code :	YZ4
Project title :	Investigating Crypto and Telecom Fraud Patterns in Hong Kong Messaging Communities
No. of students :	2
Description of project :	Hong Kong has experienced a sharp rise in cryptocurrency and telecom fraud cases conducted through messaging platforms such as WhatsApp and Telegram. Common schemes include fake investment groups, fraudulent OTC trading, romance scams leading to crypto investments, fake airdrop or DApp phishing, and impersonation of official customer service. Despite repeated warnings from the Hong Kong Police Force, SFC, and HKMA, there is a lack of systematic academic research characterizing these fraud operations. This project collects data from public sources including the Hong Kong Police CyberDefender "Scameter" database, SFC suspicious platform alerts, public Telegram groups, and news reports, subject to CUHK IRB ethics approval. The study will produce a fraud pattern taxonomy specific to the Hong Kong context, extract automatically identifiable signal features from message content and group metadata, develop a prototype detection tool leveraging LLM-based classification via prompt engineering, and deliver policy recommendations for regulators and platform operators.
Industrial collaboration :	No